



វិទ្យាស្ថានបច្ចេកវិទ្យាឌីជីថល
Institute of Digital Technology

2nd Capstone Follow-up First Version

Student Name: : Choeng Rayu IDTB100252

: Tep Somnang IDTB100190

: Tet Elite IDTB100227

: Lon Mengheng IDTB100003

: Sophal Taingchhay IDTB100086

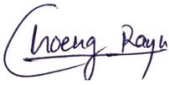
: May Kunaphivath IDTB100032

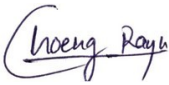
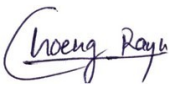
Advisor Name: Dr. Duch Dynil

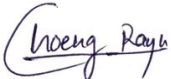
Group : 30

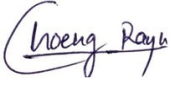
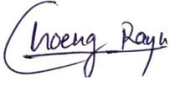
Start Date: Jan 12, 2026

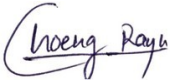
End Date: April 1, 2026

Timeline	What Students Should Do?	Missing Elements / Notes	Advisor Signature	Student Signature
First week of capstone	First Meeting: <i>allocated specific roles and responsibilities, established the project structure, and prepared the database to present our preliminary progress to the advisor</i>	Date: 09 February 2026 <i>Initially, users will manually enter prescriptions, and access to the OCR scanning feature will be limited to subscribed users. Secure communication between the frontend and backend will be implemented using encryption and password hashing to protect sensitive data. These measures will guide the project's development and security design.</i>		

Within the first 2 weeks of Capstone	Second Meeting: <i>progress of project design, requirement gathering, and technical plan. and any concerns they have faced.</i> <i>Implement project code such as;</i> <i>The first one is OCR scanning using Tesseract. Using sample data and image for testing OCR scanning to measure accuracy. Along with OCR, there is an implementation of AI using Ollama to enhance the text extract from OCR scanning.</i> <i>The second one is backend development. API design for authentication, prescription creation, family connection, medication reminder.</i> <i>The last is frontend implementation. Authentication screen design, home page screen, doctor dashboard screen are already implemented.</i>	Date: 13 February 2026 <i>Integration between backend and frontend is in progress, it is not done yet. Accelerate this action as soon as possible for complete application.</i> <i>The accuracy of OCR scanning with AI enhancement needed to be improved from 80-90 percent.</i> <i>For next meeting ,there will be successful connectivity of backend and frontend</i>		
By the end of first month	Third Meeting: <i>Application demonstration and business model proposed. Feedback on advisors and direction of improvement.</i> <i>Test accomplished connectivity between back and frontend services.</i> <i>Demonstrate the entire application processing with uploading images to OCR scanner, artificial intelligence to extract the text, and the structured presentation of the medical information in the mobile app.</i> <i>Current user verification and service assimilation.</i> <i>Propose business model and pricing of the application.</i>	Date: 20 February 2026 <i>connection between back-end and front-end with integration of the workflow was demonstrated.</i> <i>Advisor recommended switching to a system of pricing based on storage to one based on features (freemium, i.e. storage is free, but other features are paid) like access to family members.</i> <i>Simple and suggested QR code payment integration.</i> <i>The reminder scheduling should be computed dynamically and dependent upon the dosage and frequency.</i>		

		<i>Insert verification preview and save prescription information.</i> <i>Concentrate now on development of features, stability in the system, simplified authentication of phone number, and error-remedy in the next demo.</i>		
By the halfway of capstone	Fourth Meeting: Fix the notification system - Ensure alerts work correctly, handle AM/PM, default times (07:00 morning, 12:00 afternoon), and start-time logic for medicine reminders Complete API integration - Debug backend/frontend connection issues affecting notifications or other features. Improve payment system - Continue working on bank deep link integration and confirm secure payment workflow. Improve OCR accuracy - Investigate why the Kiri model produces errors. - Consider splitting images for better extraction - Reassess the need to capture patient names.	Date: 6 March 2026 Payment: Demoed basic process; screenshot-based confirmation works; deep link payment not yet functional; bank payment security implemented. Family Connection: Demoed; now works wirelessly via backend. OCR: Switched to Kiri model; text extraction works but accuracy decreased. AI / API: Gemini API allows ~500 requests/day; AI processing for OCR not yet implemented. Notifications: Not completed; may have API-related issues. Advisor Feedback & Advice: Family Connection: Investigate why two phones cannot connect; check if default limitation. Payment: Clarify payment method changes and service timing; continue deep link integration. OCR Accuracy: Focus on capturing only required info; reconsider patient name capture; consider splitting images for better accuracy. AI Strategy: Structure/clean OCR data before AI analysis for better reliability. Notifications (High Priority): Fix alert logic (AM/PM,		

	5. Finish remaining UI features ○ Ensure the app is fully functional for users. 6. Prepare AI integration ○ Structure OCR data first; implement AI-based corrections later to improve data reliability.	default times 7:00/12:00, start time alerts). ● Development Priorities: 1. Complete notification system 2. Ensure API integration 3. Finish remaining UI 4. Improve OCR 5. Implement AI improvements later		
By the last meeting of capstone	Fifth Meeting: We demonstrated the nearly completed application to the advisor. The system integration between frontend, backend, API, and database is mostly functional. Core features such as OCR prescription scanning, family connection, medication reminders, and payment workflow were presented during the demo. Overall, the system works well, although some cases still require improvement and debugging.	Date: 13 March 2026 Advisor provided several feedback suggestions for improvement during the meeting. He asked the team to measure and report the OCR letter recognition accuracy percentage to evaluate the system's performance. And also pointed out that the age extracted from the OCR can sometimes be incorrect and need to be fixed to make sure it would be more accurate. And the ui should be fix and improve to ensure the application is clear and user friendly, also about the notification system should be clearly visible and respond quickly during the demo so the functionality can be demonstrated properly. The last is that the project report is the highest priority. The report should follow all feedback and be completed by Monday night or Tuesday morning, and submitted to the advisor at least 4 or 5 days before the deadline. Slides can be prepared as well.		
By the last two weeks of capstone	Sixth Meeting: During this period, although no formal meeting was conducted, the team ensured that all work was properly finalized and reviewed. The report and presentation slides were submitted to the advisor for checking and feedback. Based on the advisor's comments, necessary improvements were	Date: 17 March 2026 ● Submitted report and slides to advisor for review ● Received detailed feedback from advisor for improvement ● Improved abstract by ensuring both Khmer and English versions clearly summarize		

	<i>made to enhance the quality and ensure timely submission.</i>	<i>technical outcomes and social impact</i> • <i>Identified and fixed inconsistencies in pricing tables and LLaMA model versions</i> • <i>Added improvements to implementation section (hardware requirements and performance considerations)</i> • <i>Enhanced system testing by including usability considerations for target users (elderly)</i> • <i>Expanded list of abbreviations for better understanding by non-technical readers</i> • <i>Improved OCR section by adding error analysis for Khmer vs English accuracy</i> • <i>Added example prompt for OCR-to-JSON transformation</i> • <i>Ensured UML diagrams are high-resolution for final submission</i> • <i>Revised and finalized report and presentation slides</i>		
Before Defense	Additional Meeting: <i>During this period, the team will conduct a final rehearsal session to prepare for the defense. The project presentation will be delivered as a full simulation of the actual defense, and feedback from the advisor will be used to refine content, delivery, and overall performance to ensure readiness.</i>	Date: 30 March 2026 • <i>Final preparation before official defense day</i> • <i>Present complete project including system, features, and results</i> • <i>Receive final feedback from advisor on content and delivery</i> • <i>Improve presentation clarity, confidence, and timing</i> • <i>Adjust slides based on feedback (content, design, flow)</i> • <i>Practice answering possible questions from the committee</i> • <i>Identify weak points and refine explanations</i> • <i>Ensure all team members are well-prepared for their parts</i>		

--	--	--	--	--

*** NOTE:**

- *This timeline serves as a guidance for students regarding their meetings, but students may meet with their advisor more frequently if needed.*
- *Students must submit both the first and final drafts of their report for their advisor's review. The same submission process applies to the slide presentation.*
- *The submission deadline is subject to school instructions. Students must ensure all documents are carefully reviewed by advisors before submitting them to the school. Using e-signatures without the advisor's consent will result in a penalty.*
- *This form encourages students to take initiative and diligently polish and update their work. Any negative grades received due to lack of diligence are the student's responsibility.*